

MATH 3260 - Chapter 6 Notes: Linear Transformations

Feedback-Revised Student Notes

Kennesaw State University - MATH 3260

Fall 2026 revision

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These notes are synchronized with the feedback-revised student notebook. They emphasize definitions, theorems, worked mathematics, conceptual connections, and interpretation while omitting notebook-specific code and worksheet instructions.

Chapter overview

1. **Section 6.1:** Introduction to Linear Transformations
2. **Section 6.2:** Kernel and Range
3. **Section 6.3:** Matrices for Linear Transformations
4. **Section 6.4:** Transition Matrices and Similarity
5. **Section 6.5:** Applications of Linear Transformations

Each section ends with a terminology table, a worksheet, and complete instructor solutions.

Original-example policy

All worked examples and numerical values are original. Themes are used selectively from the San Antonio Spurs, hiking, badminton, driving, travel, Backstreet Boys, and R&B music. Any routes, listening profiles, scores, or geometric models are fictional teaching contexts unless explicitly identified otherwise.

Section 6.1 - Introduction to Linear Transformations

Learning objectives

Students should be able to:

- identify the domain, codomain, image, preimage, and range of a transformation;
- test whether a mapping is linear;
- use the structural properties of linear transformations;
- determine a transformation from its values on a basis;
- recognize matrix-defined transformations and calculus operators as linear.

Function terminology

A transformation $T : V \rightarrow W$ assigns each vector in the domain V exactly one vector in the codomain W . If $T(\mathbf{v}) = \mathbf{w}$, then $\mathbf{w}$ is the image of $\mathbf{v}$, and $\mathbf{v}$ is a preimage of $\mathbf{w}$.

A travel-profile transformation

Define

$$T : \mathbb{R}^2 \rightarrow \mathbb{R}^2, \quad T(x, y) = (2x - y, x + 3y).$$

For the fictional travel profile $(3, 1)$,

$$T(3, 1) = (2(3) - 1, 3 + 3(1)) = (5, 6).$$

To find the preimage of $(5, 6)$, solve

$$\begin{aligned}2x - y &= 5, \\x + 3y &= 6.\end{aligned}$$

From the first equation, $y = 2x - 5$. Substitution gives

$$x + 3(2x - 5) = 6 \implies 7x = 21 \implies x = 3,$$

and therefore $y = 1$. The preimage is the single vector $(3, 1)$.

Definition: linear transformation

A mapping $T : V \rightarrow W$ is linear when $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ and $T(c\mathbf{u}) = cT(\mathbf{u})$ for all vectors and all scalars.

Verifying linearity

Let

$$T(x, y) = (2x - y, x + 3y).$$

For $\mathbf{u} = (u_1, u_2)$ and $\mathbf{v} = (v_1, v_2)$,

$$\begin{aligned}T(\mathbf{u} + \mathbf{v}) &= T(u_1 + v_1, u_2 + v_2) \\&= (2u_1 + 2v_1 - u_2 - v_2, u_1 + v_1 + 3u_2 + 3v_2) \\&= (2u_1 - u_2, u_1 + 3u_2) + (2v_1 - v_2, v_1 + 3v_2) \\&= T(\mathbf{u}) + T(\mathbf{v}).\end{aligned}$$

Also,

$$T(c\mathbf{u}) = T(cu_1, cu_2) = (2cu_1 - cu_2, cu_1 + 3cu_2) = cT(\mathbf{u}).$$

Therefore, T is linear.

Fast nonlinearity test

Every linear transformation satisfies $T(\mathbf{0}) = \mathbf{0}$. Therefore, any mapping with $T(\mathbf{0}) \neq \mathbf{0}$ is immediately non-linear.

For example,

$$F(x, y) = (x + 1, y)$$

is not linear because

$$F(0, 0) = (1, 0) \neq (0, 0).$$

A transformation determined by basis images

Suppose a linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ satisfies

$$T(\mathbf{e}_1) = (9, 1), \quad T(\mathbf{e}_2) = (20, -3), \quad T(\mathbf{e}_3) = (21, 4).$$

Because

$$(2, -1, 3) = 2\mathbf{e}_1 - \mathbf{e}_2 + 3\mathbf{e}_3,$$

linearity gives

$$\begin{aligned} T(2, -1, 3) &= 2T(\mathbf{e}_1) - T(\mathbf{e}_2) + 3T(\mathbf{e}_3) \\ &= 2(9, 1) - (20, -3) + 3(21, 4) \\ &= (61, 17). \end{aligned}$$

If T is linear and $\mathbf{x} = x_1\mathbf{e}_1 + \cdots + x_n\mathbf{e}_n$, then

$$T(\mathbf{x}) = x_1T(\mathbf{e}_1) + \cdots + x_nT(\mathbf{e}_n).$$

So knowing the basis images determines T on every vector.

Matrix-defined transformations

Every $m \times n$ matrix A defines a linear transformation

$$T : \mathbb{R}^n \rightarrow \mathbb{R}^m, \quad T(\mathbf{x}) = A\mathbf{x}.$$

For

$$A = \begin{bmatrix} 1 & 2 \\ 3 & -1 \\ 0 & 4 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 2 \\ -1 \end{bmatrix},$$

we obtain

$$T(\mathbf{v}) = A\mathbf{v} = \begin{bmatrix} 1(2) + 2(-1) \\ 3(2) - (-1) \\ 4(-1) \end{bmatrix} = \begin{bmatrix} 0 \\ 7 \\ -4 \end{bmatrix}.$$

Calculus operators

The differential operator

$$D : P_3 \rightarrow P_2, \quad D(p) = p'$$

is linear because differentiation preserves sums and scalar multiples.

For

$$p(x) = 4 - 3x + 2x^2 + x^3,$$

$$D(p) = -3 + 4x + 3x^2.$$

The definite-integral functional

$$L(p) = \int_0^1 p(x) dx$$

is also linear. For $p(x) = 1 + 2x + x^2$,

$$L(p) = 1 + 1 + \frac{1}{3} = \frac{7}{3}.$$

Closing note

A linear transformation is completely controlled by what it does to a basis. This principle leads directly to matrix representations in Section 6.3.

Definitions and terminology

Term	Meaning
Transformation	Function between vector spaces.
Domain	Set of allowable input vectors.
Codomain	Declared target vector space.
Image	Output $T(\mathbf{v})$ of a vector.
Preimage	Input vector mapped to a specified output.
Range	Set of all images of vectors in the domain.
Linear transformation	Mapping preserving addition and scalar multiplication.
Zero transformation	Transformation mapping every vector to zero.
Differential operator	Linear mapping sending a function to its derivative.

Section 6.2 - Kernel and Range of a Linear Transformation

Learning objectives

Students should be able to:

- find and interpret the kernel of a linear transformation;
- find a basis for the range;
- compute rank and nullity;
- apply the rank-nullity theorem;
- determine whether a transformation is one-to-one or onto;
- recognize isomorphic finite-dimensional vector spaces.

Where kernel and range live

For $T : V \rightarrow W$,

$$\ker(T) \subseteq V, \quad \text{Range}(T) \subseteq W.$$

The kernel is a subspace of the domain; the range is a subspace of the codomain.

Kernel and range

For $T : V \rightarrow W$, the kernel is $\ker(T) = \{\mathbf{v} \in V : T(\mathbf{v}) = \mathbf{0}\}$. The range is $\text{range}(T) = \{T(\mathbf{v}) : \mathbf{v} \in V\}$. Both are subspaces.

Projection onto the xy -plane

Consider

$$T(x, y, z) = (x, y, 0).$$

The kernel consists of vectors satisfying

$$T(x, y, z) = (0, 0, 0),$$

so $x = 0$ and $y = 0$, while z is free. Therefore,

$$\ker(T) = \{(0, 0, z) : z \in \mathbb{R}\} = \text{span}\{(0, 0, 1)\}.$$

The range is the xy -plane:

$$\text{range}(T) = \text{span}\{(1, 0, 0), (0, 1, 0)\}.$$

A rank-nullity example

Let $T : \mathbb{R}^4 \rightarrow \mathbb{R}^3$ be defined by $T(\mathbf{x}) = A\mathbf{x}$, where

$$A = \begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & -1 & 2 \\ 1 & 1 & 0 & 1 \end{bmatrix}.$$

Row reduction gives

$$\begin{aligned} \begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & -1 & 2 \\ 1 & 1 & 0 & 1 \end{bmatrix} &\xrightarrow{R_3 \leftarrow R_3 - R_1} \begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & -1 & 2 \\ 0 & 1 & -2 & 0 \end{bmatrix} \\ &\xrightarrow{R_3 \leftarrow R_3 - R_2} \begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & -1 & -2 \end{bmatrix}. \end{aligned}$$

There are three pivots, so

$$\text{rank}(T) = 3.$$

To find the kernel, solve

$$\begin{aligned} x_1 + 2x_3 + x_4 &= 0, \\ x_2 - x_3 + 2x_4 &= 0, \\ -x_3 - 2x_4 &= 0. \end{aligned}$$

Let $x_4 = t$. Then

$$x_3 = -2t, \quad x_2 = -4t, \quad x_1 = 3t.$$

Thus,

$$\ker(T) = \text{span}\{(3, -4, -2, 1)\},$$

and

$$\text{nullity}(T) = 1.$$

The rank-nullity theorem checks:

$$3 + 1 = 4 = \dim(\mathbb{R}^4).$$

Rank-nullity theorem

If $T : V \rightarrow W$ is linear and V is finite dimensional, then $\text{rank}(T) + \text{nullity}(T) = \dim(V)$.

One-to-one and onto

A linear transformation is one-to-one exactly when

$$\ker(T) = \{\mathbf{0}\}.$$

It is onto exactly when

$$\text{rank}(T) = \dim(W).$$

For the 3×4 matrix above, the rank is 3, equal to the dimension of the codomain $\mathbb{R}^3$. Therefore, T is onto. Its nullity is 1, so it is not one-to-one.

To test one-to-one, solve $T(\mathbf{v}) = \mathbf{0}$: the kernel must be $\{\mathbf{0}\}$. To test onto, verify that every codomain vector can be reached; for a matrix, this means a pivot in every row.

Isomorphism example

Define the coordinate map

$$C : P_2 \rightarrow \mathbb{R}^3, \quad C(a + bx + cx^2) = (a, b, c).$$

This mapping is linear, one-to-one, and onto. Its inverse is

$$C^{-1}(a, b, c) = a + bx + cx^2.$$

Therefore, P_2 and $\mathbb{R}^3$ are isomorphic.

Closing note

The kernel measures information lost by a transformation, while the range measures the outputs that can actually be produced.

Definitions and terminology

Term	Meaning
Kernel	Inputs mapped to the zero vector.
Range	Set of all attainable outputs.
Nullity	Dimension of the kernel.
Rank	Dimension of the range.
Rank-nullity theorem	$\text{rank}(T) + \text{nullity}(T) = \dim(V)$.
One-to-one	Distinct inputs have distinct outputs.
Onto	Every vector in the codomain is an output.
Isomorphism	Linear transformation that is one-to-one and onto.
Isomorphic spaces	Vector spaces with the same linear structure.

Section 6.3 - Matrices for Linear Transformations

Learning objectives

Students should be able to:

- construct the standard matrix of a linear transformation;
- use matrix multiplication to evaluate transformations;
- form matrices for compositions;
- find inverse transformations;
- construct matrices relative to nonstandard bases;
- represent operators between polynomial spaces.

Standard matrix theorem

If $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is linear, then its standard matrix is $A = [T(\mathbf{e}_1) \cdots T(\mathbf{e}_n)]$, and $T(\mathbf{x}) = A\mathbf{x}$.

Standard matrix from a formula

Let

$$T(x, y, z) = (x + 2y - z, 3x - y + 4z).$$

Compute the images of the standard basis vectors:

$$T(1, 0, 0) = (1, 3),$$

$$T(0, 1, 0) = (2, -1),$$

$$T(0, 0, 1) = (-1, 4).$$

Therefore,

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 3 & -1 & 4 \end{bmatrix}.$$

For $\mathbf{v} = (2, -1, 3)$,

$$T(\mathbf{v}) = A\mathbf{v} = \begin{bmatrix} 1 & 2 & -1 \\ 3 & -1 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} = \begin{bmatrix} -3 \\ 19 \end{bmatrix}.$$

After using the matrix, verify one sample vector directly from the formula for T . The formula path and matrix path must agree.

Composition

If T_1 has standard matrix A_1 and T_2 has standard matrix A_2 , then $T_2 \circ T_1$ has standard matrix A_2A_1 . The order matters.

Shear followed by rotation

Let

$$A_1 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

represent a horizontal shear, and let

$$A_2 = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

represent a counterclockwise 90° rotation.

The composition “shear first, then rotate” has matrix

$$A_2 A_1 = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 1 \end{bmatrix}.$$

The reverse composition has matrix

$$A_1 A_2 = \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix},$$

which is different.

For rotation about the z -axis, the z coordinate is unchanged, explaining the $(0, 0, 1)$ pattern. Rotations about other axes fix different coordinates. Positive orientation follows the right-hand rule.

Inverse transformation

Let

$$T(\mathbf{x}) = A\mathbf{x}, \quad A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}.$$

Because

$$\det(A) = 2(1) - 1(1) = 1 \neq 0,$$

T is invertible. The inverse matrix is

$$A^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}.$$

Thus,

$$T^{-1}(u, v) = (u - v, -u + 2v).$$

Use consistent notation:

$$T(\mathbf{x}) = A\mathbf{x} \implies T^{-1}(\mathbf{y}) = A^{-1}\mathbf{y}.$$

Here $\mathbf{y}$ is an output of T and becomes an input of T^{-1} .

A matrix relative to nonstandard bases

Let

$$T(x, y) = (x + y, 2x - y).$$

Use the domain basis

$$B = \{(1, 1), (1, -1)\}$$

and codomain basis

$$C = \{(1, 0), (1, 1)\}.$$

For the first basis vector,

$$T(1, 1) = (2, 1) = 1(1, 0) + 1(1, 1),$$

so

$$[T(1, 1)]_C = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

For the second basis vector,

$$T(1, -1) = (0, 3) = -3(1, 0) + 3(1, 1),$$

so

$$[T(1, -1)]_C = \begin{bmatrix} -3 \\ 3 \end{bmatrix}.$$

Therefore,

$$[T]_{C \leftarrow B} = \begin{bmatrix} 1 & -3 \\ 1 & 3 \end{bmatrix}.$$

A coordinate vector such as

$$[\mathbf{v}]_B = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

means $\mathbf{v} = 2\mathbf{b}_1 - \mathbf{b}_2$.

Differential operator matrix

Relative to the standard bases

$$B = \{1, x, x^2, x^3\}$$

for P_3 and

$$C = \{1, x, x^2\}$$

for P_2 , differentiation has matrix

$$[D]_{C \leftarrow B} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}.$$

Indeed, the columns are the coordinate vectors of

$$D(1) = 0, \quad D(x) = 1, \quad D(x^2) = 2x, \quad D(x^3) = 3x^2.$$

Closing note

The matrix of a transformation is a coordinate description. Changing bases changes the matrix but not the underlying transformation.

Definitions and terminology

Term	Meaning
Standard matrix	Matrix using standard bases in domain and codomain.
Composition	Applying one transformation after another.
Inverse transformation	Mapping that reverses an invertible transformation.
Transformation matrix	Matrix relative to chosen domain and codomain bases.
Coordinate vector	Coefficients of a vector relative to an ordered basis.
Differential-operator matrix	Matrix representing differentiation between polynomial spaces.

Section 6.4 - Transition Matrices and Similarity

Learning objectives

Students should be able to:

- construct transition matrices between ordered bases;
- convert coordinate vectors from one basis to another;
- relate matrices of the same transformation in different bases;
- test and use matrix similarity;
- recognize invariants preserved by similarity;
- exploit diagonal representations to compute matrix powers.

Transition matrix

If B and C are ordered bases, the transition matrix from B -coordinates to C -coordinates satisfies $[\mathbf{x}]_C = P_{C \leftarrow B}[\mathbf{x}]_B$. If the basis matrices are also called B and C , then $P_{C \leftarrow B} = C^{-1}B$.

Coordinate conversion

Let

$$B = \{(1, 1), (1, -1)\}, \quad C = \{(1, 0), (1, 1)\}.$$

The basis matrices are

$$B = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}.$$

Therefore,

$$P_{C \leftarrow B} = C^{-1}B = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 2 \\ 1 & -1 \end{bmatrix}.$$

For $\mathbf{x} = (5, 1)$,

$$[\mathbf{x}]_B = \begin{bmatrix} 3 \\ 2 \end{bmatrix}.$$

Then

$$[\mathbf{x}]_C = P_{C \leftarrow B}[\mathbf{x}]_B = \begin{bmatrix} 0 & 2 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}.$$

Similarity relation

If A is the matrix of T in one basis and A_B is the matrix in basis B , then $A_B = P^{-1}AP$. Matrices related this way are similar.

A diagonal representation

Consider

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

and the basis

$$B = \{(1, 1), (1, -1)\}.$$

The transition matrix from B to the standard basis is

$$P = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix},$$

and

$$P^{-1} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

Then

$$\begin{aligned} A_B &= P^{-1}AP \\ &= \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \\ &= \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}. \end{aligned}$$

Thus, the same transformation that mixes the standard coordinates simply scales one B -direction by 3 and the other by 1.

Similarity invariants

Similar matrices have the same:

- determinant;
- trace;
- characteristic polynomial;
- eigenvalues;
- rank.

For the matrices above,

$$\text{tr}(A) = 4 = \text{tr}(A_B),$$

and

$$\det(A) = 3 = \det(A_B).$$

Computing powers through similarity

Because

$$A = PDP^{-1}, \quad D = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix},$$

we have

$$A^n = PD^nP^{-1}.$$

Since

$$D^n = \begin{bmatrix} 3^n & 0 \\ 0 & 1 \end{bmatrix},$$

$$A^n = \frac{1}{2} \begin{bmatrix} 3^n + 1 & 3^n - 1 \\ 3^n - 1 & 3^n + 1 \end{bmatrix}.$$

For example,

$$A^5 = \begin{bmatrix} 122 & 121 \\ 121 & 122 \end{bmatrix}.$$

Closing note

Similarity separates an underlying linear transformation from the coordinate system used to describe it. A well-chosen basis can turn a complicated matrix into a simple one.

Definitions and terminology

Term	Meaning
Transition matrix	Converts coordinate vectors between ordered bases.
Basis matrix	Matrix whose columns are the basis vectors.
Similar matrices	Matrices related by $A_B = P^{-1}AP$.
Similarity invariant	Quantity unchanged by a basis change.
Diagonal representation	Matrix with only diagonal entries in a suitable basis.
Coordinate conversion	Rewriting the same vector in a different basis.

Section 6.5 - Applications of Linear Transformations

Learning objectives

Students should be able to:

- identify matrices for reflections, expansions, contractions, shears, and rotations;
- transform planar figures using matrix multiplication;
- compose geometric transformations;
- interpret determinant as area scaling;
- rotate three-dimensional objects about coordinate axes.

A determinant of 1 preserves area, but area can genuinely scale. For

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix},$$

$|\det(A)| = 6$, so every planar area is multiplied by 6.

Common planar transformations

Reflections, scalings, shears, and rotations are linear transformations of $\mathbb{R}^2$. Their matrices describe how the standard basis vectors move.

Standard 2×2 geometry matrices

Reflection in the y -axis:

$$R_y = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Reflection in the line $y = x$:

$$R_{y=x} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Horizontal scaling:

$$S_x(k) = \begin{bmatrix} k & 0 \\ 0 & 1 \end{bmatrix}.$$

Horizontal shear:

$$H(k) = \begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}.$$

Counterclockwise rotation by θ :

$$Q(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}.$$

Badminton-court shear

Model a small rectangular court outline by

$$(0, 0), (6, 0), (6, 3), (0, 3).$$

Apply the horizontal shear

$$H\left(\frac{1}{2}\right) = \begin{bmatrix} 1 & 1/2 \\ 0 & 1 \end{bmatrix}.$$

A point (x, y) becomes

$$\left(x + \frac{1}{2}y, y\right).$$

Thus, the upper corners move right by $3/2$ units, while the lower corners remain fixed.

Rotation of a travel marker

A point $(2, 0)$ rotated counterclockwise by 30° becomes

$$\begin{aligned} Q(30^\circ) \begin{bmatrix} 2 \\ 0 \end{bmatrix} &= \begin{bmatrix} \sqrt{3}/2 & -1/2 \\ 1/2 & \sqrt{3}/2 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} \sqrt{3} \\ 1 \end{bmatrix}. \end{aligned}$$

Composition and order

Suppose a fictional Spurs pennant is first stretched horizontally by a factor of 2 and then rotated 90° counterclockwise.

The scaling matrix is

$$S = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix},$$

and the rotation matrix is

$$Q = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

The combined matrix is

$$QS = \begin{bmatrix} 0 & -1 \\ 2 & 0 \end{bmatrix}.$$

The reverse order would produce

$$SQ = \begin{bmatrix} 0 & -2 \\ 1 & 0 \end{bmatrix},$$

so the two geometric results are different.

Area scaling

For a planar linear transformation with matrix A , areas are multiplied by $|\det(A)|$. A zero determinant collapses every two-dimensional region to a line or point.

For the horizontal shear

$$H(k) = \begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix},$$

$$\det(H(k)) = 1,$$

so shear preserves area.

For

$$S = \begin{bmatrix} 2 & 0 \\ 0 & 1/2 \end{bmatrix},$$

$$\det(S) = 1,$$

so this stretch-and-compress transformation also preserves area.

Three-dimensional rotation about the z -axis

A counterclockwise rotation about the z -axis has matrix

$$R_z(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

At $\theta = 90^\circ$,

$$R_z(90^\circ)(1, 2, 3) = (-2, 1, 3).$$

The z -coordinate is unchanged.

Closing note

Geometric transformations make matrix multiplication visible: each matrix moves every point, edge, and region according to the same linear rule.

Definitions and terminology

Term	Meaning
Reflection	Mirror transformation across a line or axis.
Expansion	Scaling that increases distance in a direction.
Contraction	Scaling that decreases distance in a direction.
Shear	Transformation sliding parallel layers by proportional amounts.
Rotation	Distance- and angle-preserving turn about an axis or the origin.
Area scale factor	$ \det(A) $ for a planar transformation.

Term	Meaning
Rotation matrix	Orthogonal matrix representing a rigid rotation.
Right-hand rule	Convention determining positive 3D rotation direction.

Chapter 6 synthesis

A linear transformation can be studied from several mutually reinforcing viewpoints:

formula $\longleftrightarrow$ matrix $\longleftrightarrow$ kernel and range $\longleftrightarrow$ geometry.

Changing a basis changes the matrix, but it does not change the underlying transformation. Similarity is the algebraic language for this coordinate change.

Core connections

$\ker(T)$ = information lost, $\text{range}(T)$ = outputs attainable.

$$[T]_{\text{new basis}} = P^{-1}[T]_{\text{old basis}}P.$$

composition of transformations $\longleftrightarrow$ matrix multiplication in reverse application order.

Direct check of a standard-matrix computation

Whenever a transformation is given both by a formula and by a matrix, it is useful to verify one input in both ways:

$T(\mathbf{v})$ from the defining formula,

and

$A\mathbf{v}$ from the standard matrix.

These two vectors must agree. This simple check catches sign, basis-order, and matrix-construction errors.